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Personality-Aware Multiagent Large Language Models for Strategic Interactions in Polymatrix Games

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Abstract

We present a framework that integrates game theory and psychology to model multiagent systems of large language model (LLM) agents endowed with Big Five personality traits. We embed the agents within polymatrix games and compare their strategic decisions against Nash equilibria. We found that personality traits influence strategic behavior, and that dialogue-enabled interactions achieve convergence rates of 98–100%, compared to 42–54% without dialogue. We also show how personality traits shape collective decision-making and provide insights into the behavioral realism of game-theoretic predictions in psychologically grounded AI agents.

CCS Concepts

• **Human-centered computing** → *Human computer interaction (HCI)*; • **Information systems** → **Language models**; • **Computing methodologies** → **Multi-agent systems**; **Intelligent agents**; **Natural language generation**; • **Theory of computation** → **Algorithmic game theory**; **Solution concepts in game theory**; **Network games**; • **Applied computing** → **Psychology**; **Economics**; **Sociology**.

Keywords

AI Agents, Multiagent Systems, Large Language Models, Game Theory, Nash Equilibrium, Rationality, Big Five Personality Traits

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1 Introduction

Behavioral game theory has long recognized that rationality deviates from theoretical predictions due to various psychological factors [3]. We extend this tradition by examining personality traits in strategic interactions. Large language models (LLMs) have demonstrated remarkable abilities to simulate behaviors within complex social contexts [9, 10, 14, 16, 18, 22]. They can emulate personas, enabling applications in mental health, education, and negotiation [7, 9, 15]. In this paper, we explore how synthetic personality traits affect strategic behavior in autonomous multiagent LLM systems. To this end, we introduce a framework in which LLM agents are endowed with Big Five personality profiles. The agents engage in dialogues situated within polymatrix games [4], and select strategies based on the outcomes of their interactions. This setting raises our central research question. *Do personality-conditioned dialogue actions produce systematic deviations from classical game-theoretic equilibria?* To address it, we provide (1) a multiagent framework that integrates psychological and game-theoretic modeling via natural language interactions; (2) comparisons of strategic outcomes derived from dialogue-based decisions to analytically computed Nash equilibria (NE) in polymatrix games; and (3) empirical insights into how personality traits affect rationality, offering a richer understanding of behavioral realism in game-theoretic predictions.

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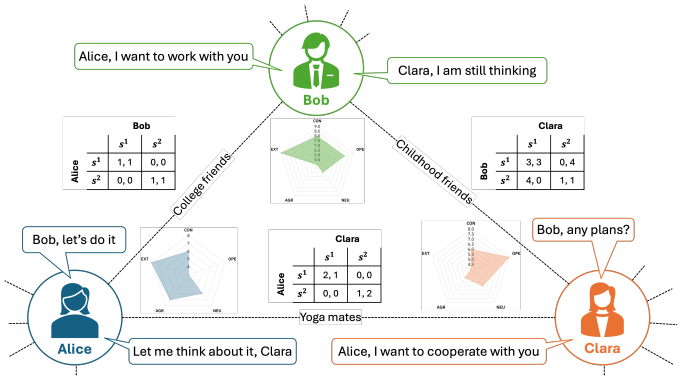

Figure 1: Example of LLM agents in a polymatrix game. Nodes represent agents with Big Five personality traits. Edges show pairwise interactions with game-theoretic payoff matrices.

2 Methodology

2.1 Multiagent LLM Framework

Our framework, illustrated in Figure 1, consists of interacting LLM agents, each having a name, age, gender, Big Five personality (ψ), and strategies with payoffs. Agents are connected in a network with edges that specify payoff matrices and relationships (e.g., friends, family, coworkers). These relationships mirror the structure of real-world social networks, where strategic interactions arise.

2.2 Personality Profiles

We adopt Big Five personality traits to define the agents along Openness to Experience (OPE), Conscientiousness (CON), Extraversion

(EXT), Agreeableness (AGR), and Neuroticism (NEU) [6, 9, 10, 12]. Trait dimension d for agent i is represented by $\psi_i^d \in \{-, +\}$, indicating low (−) or high (+) trait expression. The personality of agent i is encoded as a vector $\psi_i = (\psi_i^{OPE}, \psi_i^{CON}, \psi_i^{EXT}, \psi_i^{AGR}, \psi_i^{NEU})$.

2.3 Polymatrix Game Structure

Definition. Polymatrix games are a class of multiplayer games in which a player's payoff is determined by pairwise interactions with neighboring players [2, 4, 11]. A polymatrix game $\mathcal{G} = (V, E)$ is defined by a finite set of players $V = \{1, \ldots, n\}$, called nodes, and a finite set of edges E, taken to be unordered pairs (i, j), with $i \neq j$. Each player $i \in V$ has a finite set of strategies S_i. For each edge $(i, j) \in E$, tuple (u_{ij}, u_{ji}), where $u_{ij} : S_i \times S_j \to \mathbb{R}$ and $u_{ji} : S_j \times S_i \to \mathbb{R}$, gives the payoffs of players i and j. For player $i \in V$ and strategy profile $s = (s_1, \ldots, s_n) \in \prod_{j \in V} S_j$, the total payoff is computed as $u_i(s) = \sum_{(i,j) \in E} u_{ij}(s_i, s_j)$.

Strategy space generation. We generate a finite pure strategy set $S_i = \{s_i^1, s_i^2\}$ for each player i. The strategies are contextualized within experimental scenarios (e.g., cooperation levels, resource allocation decisions, negotiation tactics). A strategy is described in natural language as an action that an agent can adopt. This grounds the agent in natural language and guides its decision-making process regardless of how payoffs are assigned to strategies.

Payoffs of the polymatrix game. We use payoff matrices to quantify pairwise interactions in $\mathcal{G}$. While agents can express their choices in natural language, payoffs are essential for computing equilibria and comparing outcomes. We adopt 5 canonical 2×2 games that capture varying degrees of coordination, conflict, and risk, that may interact with personality traits [5]. Assigning different payoffs to the edges creates local games that model various social dynamics. Games are pure coordination (PC), Stag Hunt (SH), Prisoner's Dilemma (PD), Battle of the Sexes (BS), and Hawk-Dove (HD). The payoff matrices encode the preferences (u_{ij}, u_{ji}), and are described as follows.

$$M_{\text{PC}} = \begin{pmatrix} 1,1 & 0,0 \\ 0,0 & 1,1 \end{pmatrix}, \quad M_{\text{SH}} = \begin{pmatrix} 9,9 & 0,8 \\ 8,0 & 7,7 \end{pmatrix}, \quad M_{\text{PD}} = \begin{pmatrix} 3,3 & 0,4 \\ 4,0 & 1,1 \end{pmatrix},$$

$$M_{\text{BS}} = \begin{pmatrix} 2,1 & 0,0 \\ 0,0 & 1,2 \end{pmatrix}, \quad M_{\text{HD}} = \begin{pmatrix} 2,2 & 1,3 \\ 3,1 & 0,0 \end{pmatrix}.$$

Playing the polymatrix game. Each agent interacts randomly with its neighbors, with or without dialogue. When dialogue is not used, the agent's decisions are conditioned on the history of its past actions and those of its neighbors. When dialogue is used, each agent additionally negotiates bilaterally with its neighbors before taking an action. This setup enables a direct comparison between personality-driven dialogue and game-theoretic decisions. The resulting polymatrix game will have at least one Nash equilibrium [4, 11]. Particularly, the game may converge to a Nash equilibrium, settle on a non-equilibrium outcome, or fail to converge within a fixed cutoff of 10 rounds. Next, we examine the effects of the game type and personality traits on outcomes.

3 Experimental Results

We conducted experiments to evaluate the framework, using polymatrix games with $n \in \{3, 4, 5\}$ agents based on *Qwen2.5-72B-Instruct* [21]. For each n, 50 randomized games were generated with or without dialogue, yielding $3 \times 50 \times 2 = 300$ trials. In each trial, agents received natural language descriptions of their Big Five personality profiles, two pure strategies, and preferences.

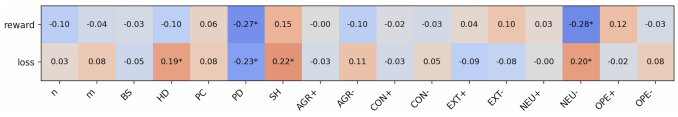

Figure 2: Correlation between game type, personality traits, and outcome measures.

We analyzed the Pearson correlations to assess how games and personality traits influence outcomes (reward and loss) across dialogue conditions. Rewards refer to the cumulative gains achieved by an agent, while losses capture regret, or suboptimal outcomes [20]. This identifies which personality dimensions and games are associated with success or failure. Figure 2 shows the relationship between input features and outcome variables across experiments. Columns are n (number of agents), m (number of neighbors), game (PC, PD, SH, BS, HD), and Big Five personality trait dimensions with positive and negative polarity (e.g., AGR+, AGR-). Colors indicate correlation coefficients, ranging from -0.2 (blue, negative correlation) to +0.2 (red, positive correlation). Values marked with an asterisk indicate $p < 0.05$. PD and SH, and some personality dimensions (e.g., NEU-) correlate with either strategic success or failure, showing differential effects of game type and personality traits on agent performance. We also looked at the effects of dialogue and group size on the convergence of strategies and alignment with Nash equilibria. Table 1 shows an improvement in both outcomes when dialogue is enabled, regardless of the number of peers (n).

Table 1: Impact of dialogue on game convergence.

Outcome Measure	Interaction	$n = 3$	$n = 4$	$n = 5$
Convergence	Without dialogue	42%	54%	48%
	With dialogue	98%	100%	100%
Nash equilibrium	Without dialogue	28%	48%	24%
	With dialogue	48%	52%	44%

4 Conclusion

Our framework offers insights into the behavioral realism of game theory and the strategic implications of personality-driven communication. LLM-based agents can embody psychological traits in strategic interactions, opening new avenues for computational social science and behavioral game theory. Future work will investigate personality adaptation, learning, and the emergence of biases [17, 19]. We emphasize the ethical dimensions of this work, noting that personality-aware agents may amplify stereotypes, reinforce power asymmetries, exhibit social identity biases, or subtly influence decision-making in consequential ways [1, 8, 13].

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References

[1] Kevin Allan, Jacobo Azcona, Somayajulu Sripada, Georgios Leontidis, Clare AM Sutherland, Louise H Phillips, and Douglas Martin. 2025. Stereotypical bias amplification and reversal in an experimental model of human interaction with generative artificial intelligence. *Royal Society Open Science* 12, 4 (2025), 241472.

[2] Yang Cai, Ozan Candogan, Constantinos Daskalakis, and Christos Papadimitriou. 2016. Zero-sum polymatrix games: A generalization of minmax. *Mathematics of Operations Research* 41, 2 (2016), 648–655.

[3] Colin Camerer. 2003. *Behavioral game theory: Experiments in strategic interaction*. Princeton university press.

[4] Argyrios Deligkas, John Fearnley, Rahul Savani, and Paul Spirakis. 2017. Computing approximate Nash equilibria in polymatrix games. *Algorithmica* 77, 2 (2017), 487–514.

[5] Robert S Gibbons. 1992. *Game theory for applied economists*. Princeton University Press.

[6] Lewis R Goldberg. 1992. The development of markers for the Big-Five factor structure. *Psychological assessment* 4, 1 (1992), 26.

[7] Y. Hicke, A. Agarwal, Q. Ma, and P. Denny. 2023. Chata: Towards an intelligent question-answer teaching assistant using open-source llms. *arXiv preprint arXiv:2311.02775* (2023).

[8] Tiancheng Hu, Yara Kyrychenko, Steve Rathje, Nigel Collier, Sander van der Linden, and Jon Roozenbeek. 2025. Generative language models exhibit social identity biases. *Nature Computational Science* 5, 1 (2025), 65–75.

[9] Yin Jou Huang and Rafik Hadfi. 2024. How Personality Traits Influence Negotiation Outcomes? A Simulation based on Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 10336–10351. doi:10.18653/v1/2024.findings-emnlp.605

[10] Yin Jou Huang and Rafik Hadfi. 2025. Beyond Self-Reports: Multi-Observer Agents for Personality Assessment in Large Language Models. *arXiv preprint arXiv:2504.08399* (2025).

[11] Elena Janovskaja. 1968. Equilibrium points in polymatrix games. *Lithuanian Mathematical Journal* 8, 2 (1968), 381–384.

[12] Oliver P John, Sanjay Srivastava, et al. 1999. The Big-Five trait taxonomy: History, measurement, and theoretical perspectives. (1999).

[13] Gulnara Z Karimova. 2025. Ethical Foundations of AI with Personality. In *Humanizing AI with Personality*. Springer, 19–31.

[14] M. Kosinski. 2023. Theory of mind may have spontaneously emerged in large language models. *CoRR* abs/2302.02083 (2023).

[15] T. Lai, Y. Shi, Z. Du, J. Wu, K. Fu, Y. Dou, and Z. Wang. 2023. Psy-llm: Scaling up global mental health psychological services with ai-based large language models. *arXiv preprint arXiv:2307.11991* (2023).

[16] A. K. Lampinen et al. 2024. Language models, like humans, show content effects on reasoning tasks. *PNAS nexus* 3, 7 (2024).

[17] Shahed Masoudian, Gustavo Escobedo, Hannah Strauss, and Markus Schedl. 2025. Investigating Gender Bias in LLM-Generated Stories via Psychological Stereotypes. *arXiv preprint arXiv:2508.03292* (2025).

[18] Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*. 1–22.

[19] Aadesh Salecha, Molly E Ireland, Shashanka Subrahmanya, João Sedoc, Lyle H Ungar, and Johannes C Eichstaedt. 2024. Large language models display human-like social desirability biases in Big Five personality surveys. *PNAS nexus* 3, 12 (2024), pgae533.

[20] Yoav Shoham and Kevin Leyton-Brown. 2008. *Multiagent systems: Algorithmic, game-theoretic, and logical foundations*. Cambridge University Press.

[21] Qwen Team. 2024. Qwen2.5: A Party of Foundation Models. https://qwenlm.github.io/blog/qwen2.5/

[22] Zekun Moore Wang, Zhongyuan Peng, Haoran Que, Jiaheng Liu, Wangchunshu Zhou, Yuhan Wu, Hongcheng Guo, Ruitong Gan, Zehao Ni, Jian Yang, et al. 2023. Rolellm: Benchmarking, eliciting, and enhancing role-playing abilities of large language models. *arXiv preprint arXiv:2310.00746* (2023).